

Functions

Data Science in a Box

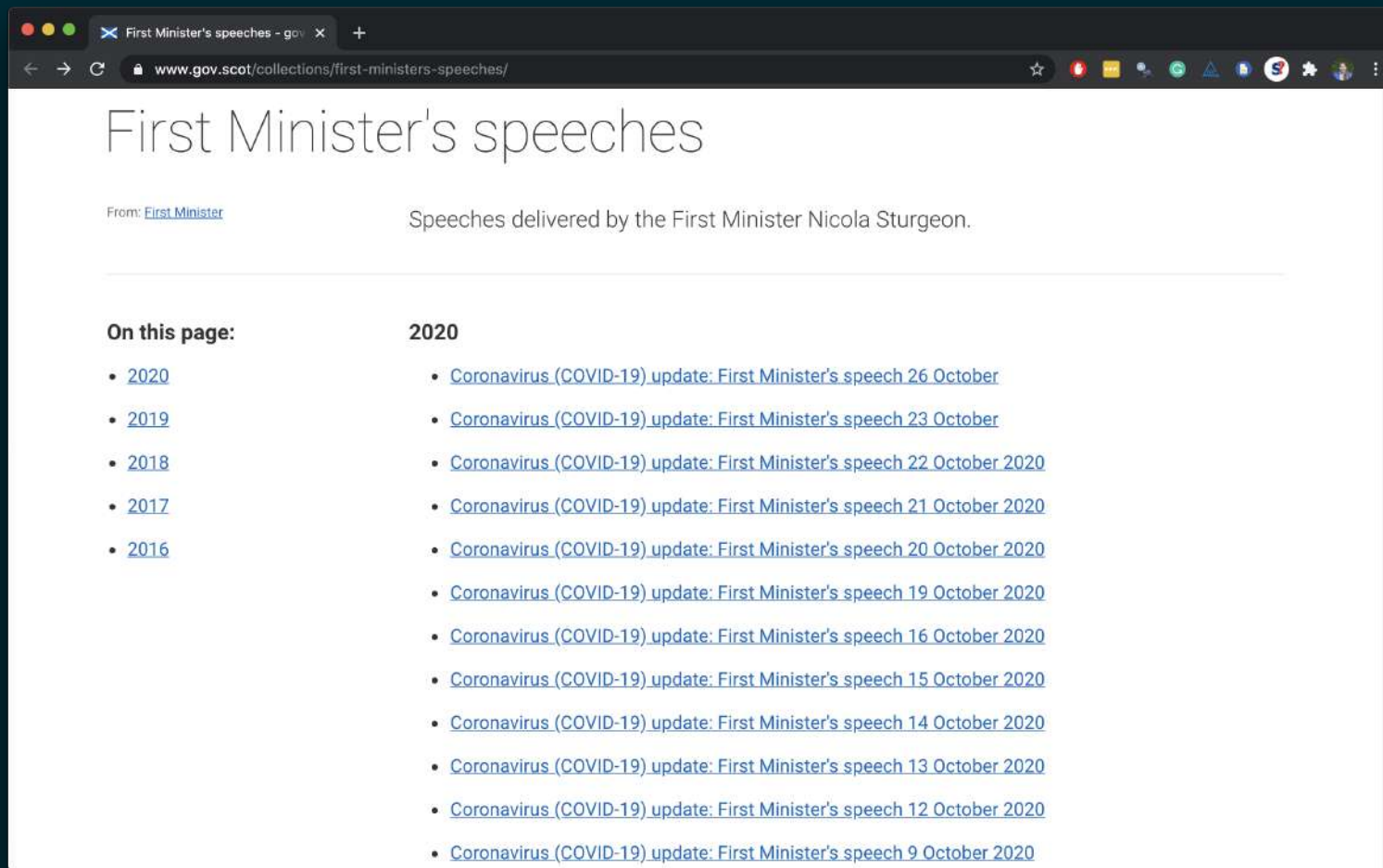
datasciencebox.org



First Minister's COVID speeches



Start with



The screenshot shows a web browser window with the address bar displaying `www.gov.scot/collections/first-ministers-speeches/`. The page title is "First Minister's speeches". Below the title, it says "From: [First Minister](#)" and "Speeches delivered by the First Minister Nicola Sturgeon."

On this page:

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2020

- [Coronavirus \(COVID-19\) update: First Minister's speech 26 October](#)
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- [Coronavirus \(COVID-19\) update: First Minister's speech 9 October 2020](#)

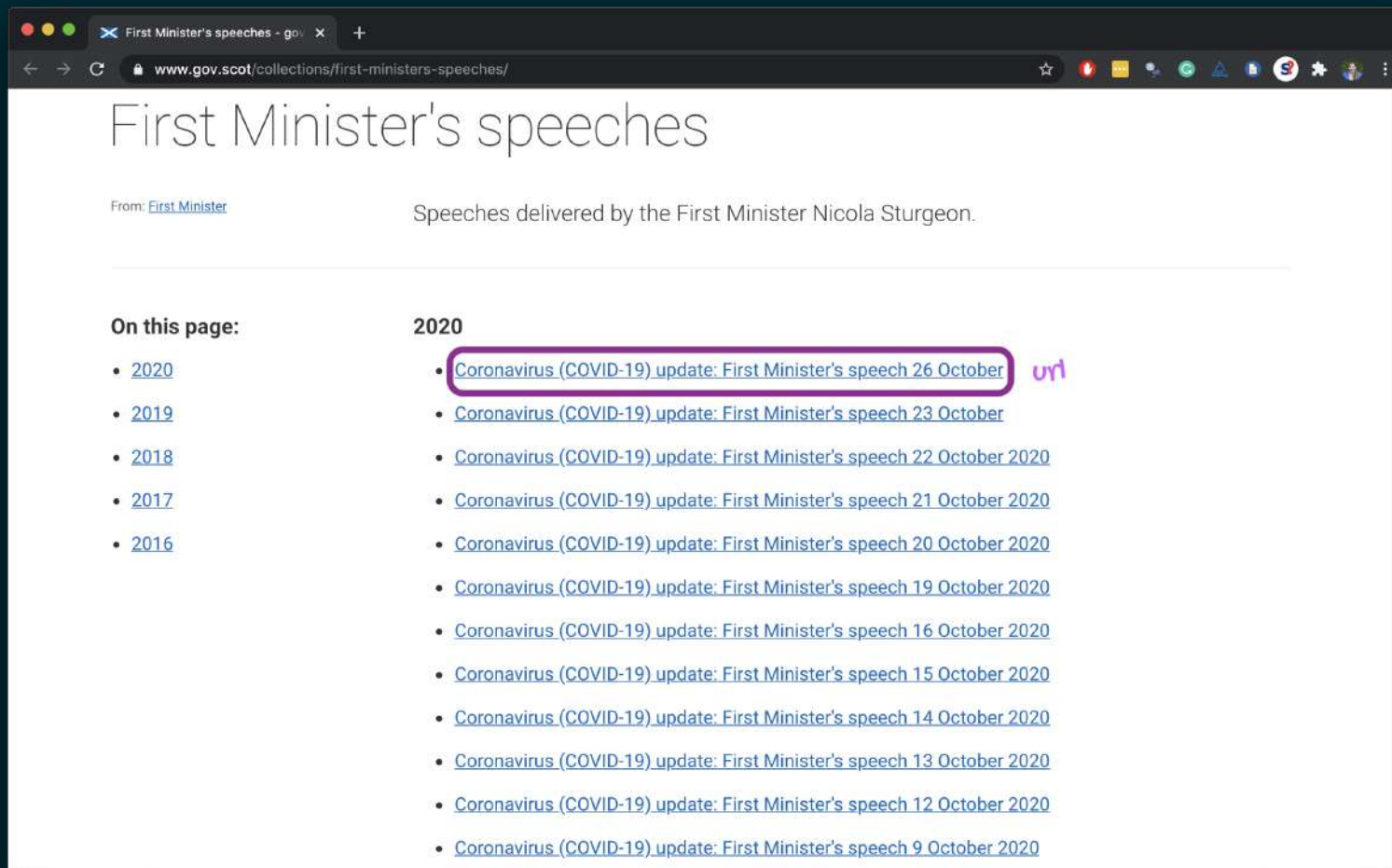


End with

```
## # A tibble: 218 × 6
##   title          date      location abstract text  url
##   <chr>         <date>    <chr>    <chr>    <chr> <chr>
## 1 Coronavirus (COVID-1... 2021-04-20 St Andr... Stateme... "Goo... http...
## 2 Coronavirus (COVID-1... 2021-04-13 St Andr... Stateme... "Tha... http...
## 3 Coronavirus (COVID-1... 2021-04-06 St Andr... Stateme... "Goo... http...
## 4 Coronavirus (COVID-1... 2021-03-30 St Andr... Stateme... "Tha... http...
## 5 Coronavirus (COVID-1... 2021-03-24 Scottis... Stateme... "Tha... http...
## 6 Coronavirus (Covid-1... 2021-03-23 The Sco... Stateme... "Pre... http...
## 7 Coronavirus (COVID-1... 2021-03-18 Scottis... Stateme... "Tha... http...
## 8 Coronavirus (COVID-1... 2021-03-17 St Andr... Stateme... "\nG... http...
## 9 Coronavirus (COVID-1... 2021-03-16 Scottis... Stateme... "Pre... http...
## 10 Coronavirus (COVID-1... 2021-03-15 St Andr... Stateme... "\nG... http...
## 11 Coronavirus (COVID-1... 2021-03-11 Scottis... Stateme... "I c... http...
## 12 Coronavirus (COVID-1... 2021-03-09 Scottis... Stateme... "Pre... http...
## 13 Coronavirus (COVID-1... 2021-03-05 Scottis... Parliam... "Hel... http...
## 14 Coronavirus (COVID-1... 2021-03-04 Scottis... Parliam... "I w... http...
## 15 Coronavirus (COVID-1... 2021-03-02 Scottis... Stateme... "Pre... http...
## # ... with 203 more rows
```



www.gov.scot/collections/first-ministers-speeches



The screenshot shows a web browser window with the address bar displaying 'www.gov.scot/collections/first-ministers-speeches/'. The page title is 'First Minister's speeches'. Below the title, it says 'From: [First Minister](#)' and 'Speeches delivered by the First Minister Nicola Sturgeon.'.

On this page:

- [2020](#)
- [2019](#)
- [2018](#)
- [2017](#)
- [2016](#)

2020

- [Coronavirus \(COVID-19\) update: First Minister's speech 26 October](#) *url*
- [Coronavirus \(COVID-19\) update: First Minister's speech 23 October](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 22 October 2020](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 21 October 2020](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 20 October 2020](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 19 October 2020](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 16 October 2020](#)
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- [Coronavirus \(COVID-19\) update: First Minister's speech 9 October 2020](#)



Coronavirus (COVID-19) update: First Minister's speech 26 October

Published: 26 Oct 2020
From: First Minister
Part of: Coronavirus in Scotland: Public safety and emergencies
Delivered by: First Minister Nicola Sturgeon
Location: St Andrew's House, Edinburgh

Statement given by First Minister Nicola Sturgeon at a media briefing in St Andrew's House on Monday 26 October 2020.

This document is part of a collection

Coronavirus update from the First Minister: 26 October 2020

Stick with it.
For yourselves and each other.

CORONAVIRUS UPDATE

Press conference
26 October 2020

Good afternoon, and thanks for joining us. I want to start with the usual daily report on the COVID statistics.

The total number of positive cases reported yesterday was 1,122.

This represents 7.1% of the total number of tests carried out. 428 of the new cases were in Greater Glasgow and Clyde, 274 in Lanarkshire, 105 in Lothian and



Plan

1. Scrape `title`, `date`, `location`, `abstract`, and `text` from a few COVID-19 speech pages to develop the code
2. Write a function that scrapes `title`, `date`, `location`, `abstract`, and `text` from COVID-19 speech pages
3. Scrape the `urls` of COVID-19 speeches from the main page
4. Use this function to scrape from each individual COVID-19 speech from these `urls` and create a data frame with the columns `title`, `date`, `location`, `abstract`, `text`, and `url`



Scrape data from a few COVID-19 speech pages

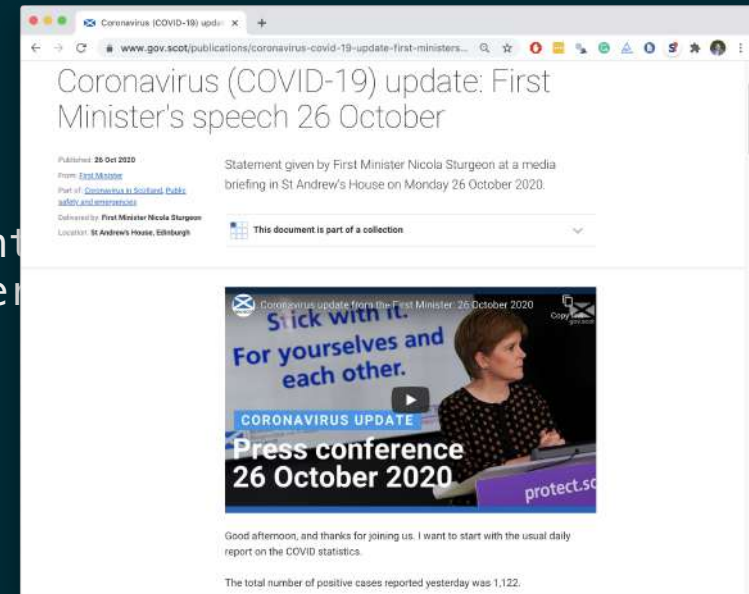


Read page for 26 Oct speech

```
url <- "https://www.gov.scot/publications/coronavirus-covid-19-update-first-ministers-speech/speech_page"
speech_page <- read_html(url)
```

speech_page

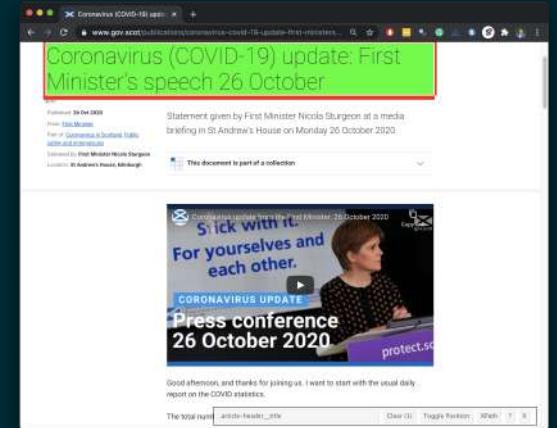
```
## {html_document}
## <html dir="ltr" lang="en">
## [1] <head>\n<meta http-equiv="Content-Type" content="text/html">\n<title>Coronavirus (COVID-19) update: First Minister's speech 26 October</title>\n</head>\n## [2] <body class="fontawesome site-header__container">
```



Extract title

```
title <- speech_page %>%  
  html_node(".article-header__title") %>%  
  html_text()  
  
title
```

```
## [1] "Coronavirus (COVID-19) update: First Minister's speech 26 October"
```



Extract date

```
library(lubridate)

speech_page %>%
  html_node(".content-data__list:nth-child(1) strong") %>%
  html_text()
```

```
## [1] "26 Oct 2020"
```

```
date <- speech_page %>%
  html_node(".content-data__list:nth-child(1) strong") %>%
  html_text() %>%
  dmy()
date
```

```
## [1] "2020-10-26"
```



Extract location

```
location <- speech_page %>%  
  html_node(".content-data__list+ .content-data__list strong")  
  html_text()
```

location

```
## [1] "St Andrew's House, Edinburgh"
```



Extract abstract

```
abstract <- speech_page %>%  
  html_node(".leader--first-para p") %>%  
  html_text()
```

```
abstract
```

```
## [1] "Statement given by First Minister Nicola Sturgeon at a media briefing at St Andrew's House on Monday 26 October 2020"
```

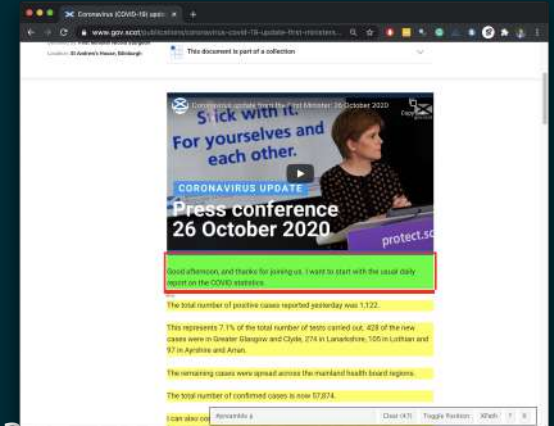


Extract text

```
text <- speech_page %>%  
  html_nodes("#preamble p") %>%  
  html_text() %>%  
  list()
```

text

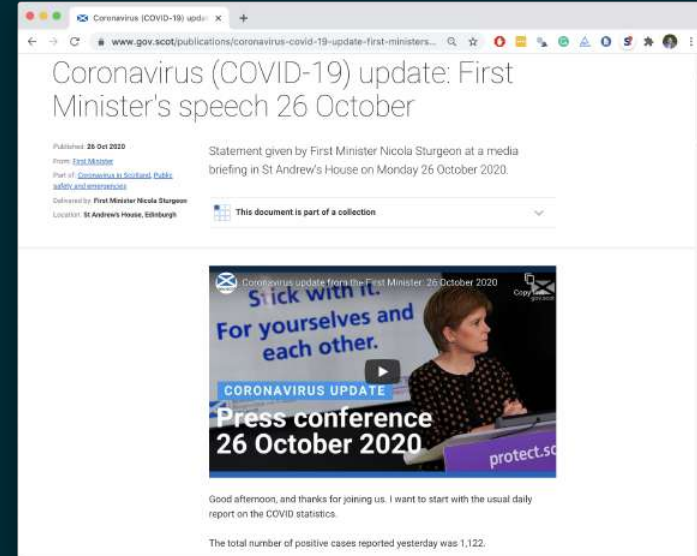
```
## [[1]]  
## [1] "\nGood afternoon, and thanks for joining us. I want to start with the usual daily report  
## [2] "The total number of positive cases reported yesterday was 1,122."  
## [3] "This represents 7.1% of the total number of tests carried out. 428 of the new cases were  
## [4] "The remaining cases were spread across the mainland health board regions.&nbsp;"  
## [5] "The total number of confirmed cases is now 57,874."  
## [6] "I can also confirm that 1,152 people are in hospital – that is an increase of 36 from  
## [7] "90 people are in intensive care, which is four more than yesterday."  
## [8] "And I regret to say that in the last 24 hours, one further death has been registered on  
## [9] "We also reported 11 deaths on Saturday, and one yesterday.&nbsp; So since the last briefing  
## [10] "That reminds us again of how dangerous this virus can be and I want to send my condolences  
... .
```



Put it all in a data frame

```
oct_26_speech <- tibble(  
  title      = title,  
  date       = date,  
  location   = location,  
  abstract   = abstract,  
  text       = text,  
  url        = url  
)  
  
oct_26_speech
```

```
## # A tibble: 1 × 6  
##   title                                date      location abstract text    url  
##   <chr>                                <date>    <chr>    <chr>    <lis> <chr>  
## 1 Coronavirus (COVID-19... 2020-10-26 St Andr... Stateme... <chr> http...
```



Read page for 23 Oct speech

```
url <- "https://www.gov.scot/publications/coronavirus-covid-19-update-first-ministers-speeches/23-october-2020/"
speech_page <- read_html(url)
```

```
speech_page
```

```
## {html_document}
## <html dir="ltr" lang="en">
## [1] <head>\n<meta http-equiv="Content-Type" content="text/html ...
## [2] <body class="fontawesome site-header__container">\n\n\n\n\n ...
```



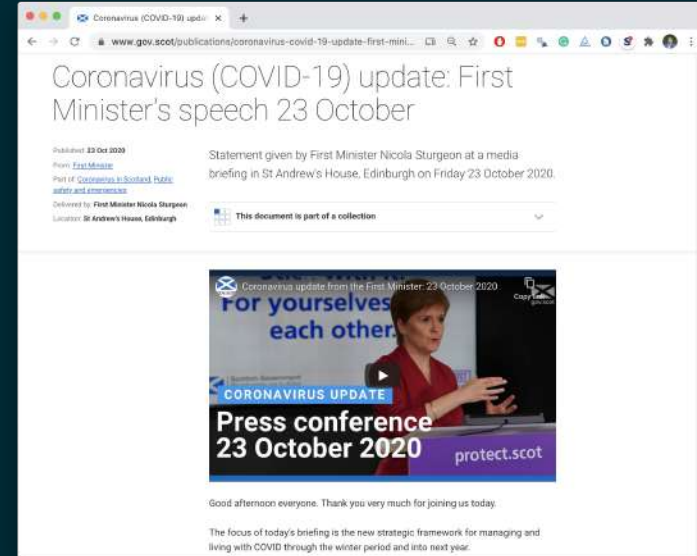
Extract components of 23 Oct speech

```
title <- speech_page %>%  
  html_node(".article-header__title") %>%  
  html_text()  
  
date <- speech_page %>%  
  html_node(".content-data__list:nth-child(1) strong") %>%  
  html_text() %>%  
  dmy()  
  
location <- speech_page %>%  
  html_node(".content-data__list+ .content-data__list strong") %>%  
  html_text()  
  
abstract <- speech_page %>%  
  html_node(".leader--first-para p") %>%  
  html_text()  
  
text <- speech_page %>%  
  html_nodes("#preamble p") %>%  
  html_text() %>%  
  list()
```



Put it all in a data frame

```
oct_23_speech <- tibble(  
  title      = title,  
  date       = date,  
  location   = location,  
  abstract   = abstract,  
  text       = text,  
  url        = url  
)  
  
oct_23_speech
```



```
## # A tibble: 1 × 6  
##   title                                date      location abstract text  url  
##   <chr>                                <date>    <chr>    <chr>    <lis> <chr>  
## 1 Coronavirus (COVID-19... 2020-10-23 St Andr... Stateme... <chr> http...
```



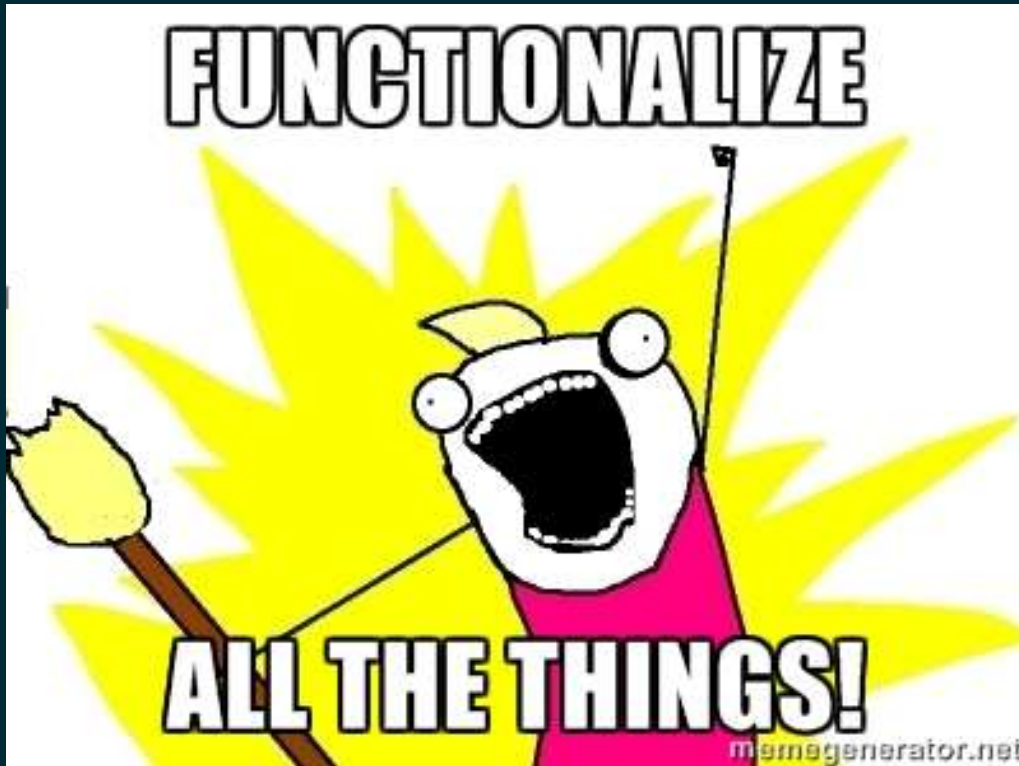
this is getting tiring...



Functions



When should you write a function?



When you've copied and pasted a block of code more than twice.

How many times will we need to copy and paste the code we developed to scrape data on all of First Minister's COVID-19 speeches?

The screenshot shows the Scottish Government website at www.gov.scot/collections/first-ministers-speeches/. A search filter for 'Coronavirus (COVID-19)' is applied, resulting in 1/142 items. A handwritten '142!' is next to the count. The page title is 'First Minister's speeches' and it is described as 'Speeches delivered by the First Minister Nicola Sturgeon'. The 'On this page:' section lists years from 2020 to 2016. The '2020' section lists 14 speeches, each with a link to the speech and a date.

Scottish Government
Riaghaltas na h-Alba
gov.scot

Coronavirus (COVID-19) 1/142 142!

Search site

About Topics News Publications Consultations Blogs

Home >

COLLECTION

First Minister's speeches

From: [First Minister](#)

Speeches delivered by the First Minister Nicola Sturgeon.

On this page:

- [2020](#)
- [2019](#)
- [2018](#)
- [2017](#)
- [2016](#)

2020

- [Coronavirus \(COVID-19\) update: First Minister's speech 26 October](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 23 October](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 22 October 2020](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 21 October 2020](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 20 October 2020](#)
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- [Coronavirus \(COVID-19\) update: First Minister's speech 16 October 2020](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 15 October 2020](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 14 October 2020](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 13 October 2020](#)
- [Coronavirus \(COVID-19\) update: First Minister's speech 12 October 2020](#)



Why functions?

- Automate common tasks in a more powerful and general way than copy-and-pasting:
 - Give your function an evocative name that makes your code easier to understand
 - As requirements change, only need to update code in one place, instead of many
 - Eliminate chance of making incidental mistakes when you copy and paste (i.e. updating a variable name in one place, but not in another)
- Down the line: Improve your reach as a data scientist by writing functions (and packages!) that others use



Assuming that the page structure is the same for each speech page, how many "things" do you need to know for each speech page to scrape the data we want from it?

```
url_23_oct <- "https://www.gov.scot/publications/coronavirus-covid-19-update-first-ministers-speech-23-oct"
speech_page <- read_html(url_23_oct)

title <- speech_page %>%
  html_node(".article-header__title") %>%
  html_text()

date <- speech_page %>%
  html_node(".content-data__list:nth-child(1) strong") %>%
  html_text() %>%
  dmy()

location <- speech_page %>%
  html_node(".content-data__list+ .content-data__list strong") %>%
  html_text()

abstract <- speech_page %>%
  html_node(".leader--first-para p") %>%
  html_text()

text <- speech_page %>%
  html_nodes("#preamble p") %>%
  html_text() %>%
  list()

tibble(
  title = title, date = date, location = location,
  abstract = abstract, text = text, url = url
```



Turn your code into a function

- Pick a short but informative **name**, preferably a verb.

```
scrape_speech <-
```



Turn your code into a function

- Pick a short but evocative **name**, preferably a verb.
- List inputs, or **arguments**, to the function inside `function`. If we had more the call would look like `function(x, y, z)`.

```
scrape_speech <- function(x){  
  
  
  
  
  
  
}
```



Turn your code into a function

- Pick a short but informative **name**, preferably a verb.
- List inputs, or **arguments**, to the function inside `function`. If we had more the call would look like `function(x, y, z)`.
- Place the **code** you have developed in body of the function, a `{` block that immediately follows `function(...)`.

```
scrape_speech <- function(url){  
  
  # code we developed earlier to scrape info  
  # on single art piece goes here  
  
}
```



scrape_speech()

```
scrape_speech <- function(url) {  
  speech_page <- read_html(url)  
  
  title <- speech_page %>%  
    html_node(".article-header__title") %>%  
    html_text()  
  
  date <- speech_page %>%  
    html_node(".content-data__list:nth-child(1) strong") %>%  
    html_text() %>%  
    dmy()  
  
  location <- speech_page %>%  
    html_node(".content-data__list+ .content-data__list strong") %>%  
    html_text()  
  
  abstract <- speech_page %>%  
    html_node(".leader--first-para p") %>%  
    html_text()  
  
  text <- speech_page %>%  
    html_nodes("#preamble p") %>%  
    html_text() %>%  
    list()  
  
  tibble(  
    title = title, date = date, location = location,  
    abstract = abstract, text = text, url = url  
  )  
}
```



Function in action

```
scrape_speech(url = "https://www.gov.scot/publications/coronavirus-covid-19-update-first-  
glimpse()")
```

```
## Rows: 1  
## Columns: 6  
## $ title      <chr> "Coronavirus (COVID-19) update: First Minister..  
## $ date       <date> 2020-10-26  
## $ location   <chr> "St Andrew's House, Edinburgh"  
## $ abstract   <chr> NA  
## $ text       <list> <"\nGood afternoon, and thanks for joining us..  
## $ url        <chr> "https://www.gov.scot/publications/coronavirus..
```



Function in action

```
scrape_speech(url = "https://www.gov.scot/publications/coronavirus-covid-19-update-first-  
glimpse()")
```

```
## Rows: 1  
## Columns: 6  
## $ title      <chr> "Coronavirus (COVID-19) update: First Minister..  
## $ date       <date> 2020-10-23  
## $ location   <chr> "St Andrew's House, Edinburgh"  
## $ abstract   <chr> NA  
## $ text       <list> <"\nGood afternoon everyone. Thank you very m..  
## $ url        <chr> "https://www.gov.scot/publications/coronavirus..
```



Function in action

```
scrape_speech(url = "https://www.gov.scot/publications/coronavirus-covid-19-update-first-glimpse()")
```

```
## Rows: 1
## Columns: 6
## $ title      <chr> "Coronavirus (COVID-19) update: First Minister...
## $ date       <date> 2020-10-22
## $ location   <chr> "St Andrew's House, Edinburgh"
## $ abstract   <chr> NA
## $ text       <list> <"\nGood afternoon, let me start as usual wit...
## $ url        <chr> "https://www.gov.scot/publications/coronavirus..."
```



Writing functions



What goes in / what comes out?

- They take input(s) defined in the function definition

```
function([inputs separated by commas]){  
  # what to do with those inputs  
}
```

- By default they return the last value computed in the function

```
scrape_page <- function(x){  
  # do bunch of stuff with the input...  
  
  # return a tibble  
  tibble(...)  
}
```

- You can define more outputs to be returned in a list as well as nice print methods (but we won't go there for now...)



What is going on here?

```
add_2 <- function(x){  
  x + 2  
  1000  
}
```

```
add_2(3)
```

```
## [1] 1000
```

```
add_2(10)
```

```
## [1] 1000
```



Naming functions

"There are only two hard things in Computer Science: cache invalidation and naming things." - Phil Karlton



Naming functions

- Names should be short but clearly evoke what the function does
- Names should be verbs, not nouns
- Multi-word names should be separated by underscores (`snake_case` as opposed to `camelCase`)
- A family of functions should be named similarly (`scrape_page()`, `scrape_speech()` OR `str_remove()`, `str_replace()` etc.)
- Avoid overwriting existing (especially widely used) functions

```
# JUST DON'T  
mean <- function(x){  
  x * 3  
}
```

